

Quantum key distribution

Instructions for laboratory assignment 4 in TSIT03 Cryptology
Jan-Åke Larsson, ISY, LiU

September 30, 2025

Before you start the assignment, make sure that you have read and understood the basics about quantum key distribution (QKD or “quantum cryptography”) that are mentioned in this document. Also, look at the slides from the lecture. You need to know what quantum key distribution is meant to do (and what it is not supposed to do).

In the following pages we include parts of a lab manual for the equipment. The full manual is for a course on Quantum communication (see e.g., <https://studieinfo.liu.se/kurs/tsit13>), so some things are left out that one would need when building and connecting the sender, receiver, and eavesdropping station. In this lab you will use the equipment rather than build it. We have also crossed out some errors and things you will not need, for example: ~~text you don't need~~ and added some text in blue frames.

1 Preparation tasks

Read Chapter 5 and 6 to familiarize yourself with the equipment. Continue to read the exercises in chapter 8. If you have any questions about the exercises, please contact the lab-assistant. During the lab you will need to generate several random 25-bit strings, one of which is the eavesdropper Eve's guess on which basis that is used for each transmitted bit (in experiment 2). Below there are two short preparation tasks.

Task: Generate random bits for the basis guess uniformly at random.

Question: Is generating the basis choice uniformly at random always the best way for the eavesdropper to guess the basis choice?

Alice and Bob will have to generate three random bit strings for each experiment, two for the basis choice at Alice and Bob respectively, and one for the initial bits to be transmitted from Alice

to Bob. These random bit strings can be generated during the lab and kept secret to Alice and Bob station respectively. By hiding the basis choice, the bits that Alice will transmit, and Bob's basis choice, you can see for yourself that the BB84 protocol for QKD works and the two parties will get the same secret key without communicating it directly.

2 During the lab

Do the exercises in Chapter 8 and note the output data in the tables in Chapter 9. An important practical tip is to use the receiver station on the Eve platform as the legitimate receiver Bob in the first exercises (1-7) to ensure that the equipment is calibrated correctly. For the subsequent exercises (8-12), where the eavesdropper should be present, use the equipment as described.



EDU-QCRY1
EDU-QCRY1/M

Quantum Cryptography
Demonstration Kit

Manual



Chapter 5 Fundamentals of Quantum Cryptography

This section explains how quantum cryptography works and the steps required to carry it out. It starts with a brief introduction and then explains the one-time pad which turns a message and a key into an encrypted message. This is followed by the generation of the keys, constituting the essential element of quantum cryptography.

The value of quantum cryptography lies in the safety from interception. Section 5.4 discusses how an eavesdropper can be detected using this method.

5.1. Introduction

Cryptography describes the encryption of data: that is, rendering a message unrecognizable, ideally making it readable only for the sender and the recipient. This means that the encrypted message is only meaningful if the key to decode it is known. The security of the key is based either on complex underlying algorithms or on practical constraints such as the factorization of large numbers.

All classical cryptography methods have the disadvantage that one can never be sure that the key will not be "cracked" eventually. This fundamental problem however can be solved with the use of quantum physics. One of the core rules governing quantum physics is that observing the state of a photon or particle simultaneously causes the state to change. This principle, along with true random number generation, is what allows a user to generate a *random* key that is *known only to the sender and the recipient*. As an added feature, any attempt at interception can theoretically be identified.

There already exist some examples of encryption systems that employ quantum cryptography. These systems are commercially available today, for example at <http://www.idquantique.com/quantum-safe-crypto/>

5.2. The One Time Pad

The one-time pad, also known as a single-use key, is an encryption method that is 100% secure in principle, provided that all requirements are fully met. Quantum physics merely helps meet these requirements, whereas the method itself is a classical encryption technique.

Imagine an encryption key that consists entirely of a perfectly random sequence of 0s and 1s called "bits". Now imagine the message also consists of 0s and 1s. Binary addition of the message and encryption key can be performed to obtain another chain of 0s and 1s which is completely random as well. This results in the encrypted message.

The “calculation rules” that apply for binary addition are as follows:

- $0 + 0 = 0$
- $1 + 0 = 1$
- $0 + 1 = 1$
- $1 + 1 = 0$

When the intended recipient obtains the encrypted message, they will use binary addition on the encrypted message and encryption key. This will then produce the original message.

By way of an example, we can encode the word “Test”. Each letter can be translated into a five-digit binary code as shown in the table below (see Chapter 9 for a table that converts alphabet letters to binary code):

Word	T					E					S					T				
Binary word	1	0	0	1	1	0	0	1	0	0	1	0	0	1	0	1	0	0	1	1
+																				
Key (random)	1	1	0	1	0	1	0	0	0	1	1	0	1	0	0	1	1	1	0	1
Encrypted message	0	1	0	0	1	1	0	1	0	1	0	0	1	1	0	0	1	1	1	0
+																				
Key (as above)	1	1	0	1	0	1	0	0	0	1	1	0	1	0	0	1	1	1	0	1
Binary word	1	0	0	1	1	0	0	1	0	0	1	0	0	1	0	1	0	0	1	1
Word	T					E					S					T				

If the encrypted message is intercepted, the eavesdropper requires the key in order to decode it. Without the key, the random sequence of zeroes and ones produce complete “gibberish” when converted to a word. This makes the message entirely safe from interception.

To summarize the essential requirements:

1. The key has to be at least as long as the message.
2. The key must only be used once.
3. The key must be completely random.
4. The key must be known only to the sender and the recipient.

Requirement 1 is easy to meet by the sender, who can only encrypt a number of bits that is less than or equal to the number of available key bits.

Requirement 2 is the responsibility of the sender and recipient, and is easily realizable as well.

Requirement 3 is difficult to meet upon closer inspection, since every random number generator is ultimately based on an algorithm. This means that random numbers generated by a computer are always merely “pseudo-random”. However, quantum physics can be used to solve this problem since it makes true randomness possible. This is discussed in more detail in Section 5.5.

Requirement 4 is problematic as well, since the classical transmission of a key opens up the possibility of intercepting it. This problem too can be solved with quantum physics. The approach to the secret distribution of the key is discussed in the next subsection.

5.3. Key Distribution

5.3.1. $\lambda/2$ Plate and Data Transmission with One Basis

This subsection is intended to facilitate a better understanding of the experimental setup by briefly running through the process of transmitting data with one basis. Actual quantum cryptography (in the real world and in this analogy experiment) works with two bases, which is described in the next subsection.

A photon is to be used to transmit a “0” or a “1”. In this example, the polarization direction is used as the bit: A photon with horizontal polarization is interpreted as a “0”, one with vertical polarization as a “1”.

So what would an experimental setup look like that can transmit data this way? An example is shown in Figure 2.

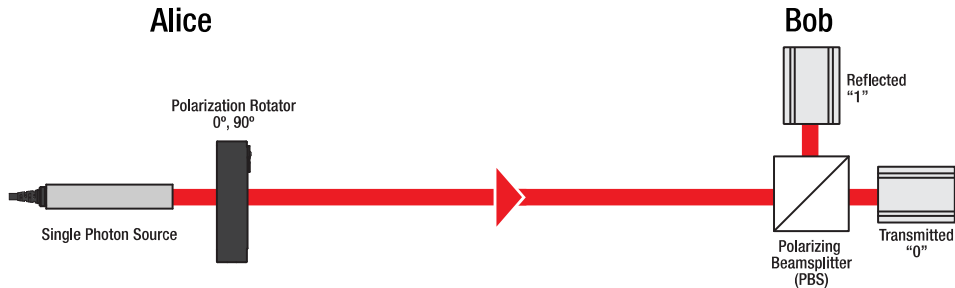


Figure 2 Data Transmission with One Polarization Basis

The sending unit “Alice” consists of a single photon source which is polarized horizontally and a $\lambda/2$ plate.

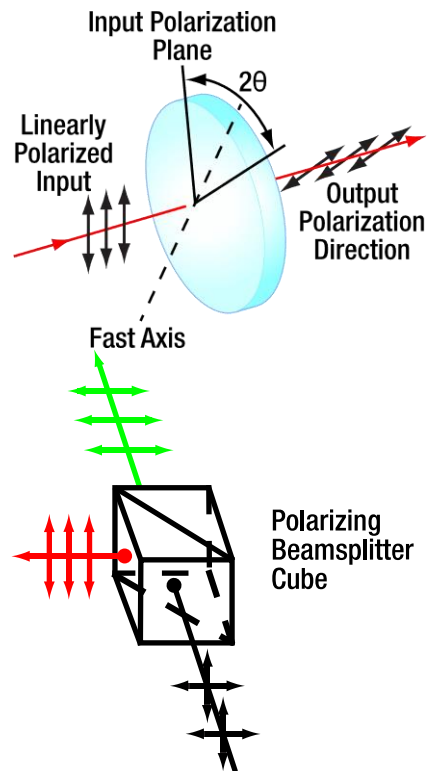
The $\lambda/2$ plate rotates the polarization of the incident light by **double** the physical rotation angle of the wave plate. For example, when the wave plate is rotated physically by 45° relative to the incoming polarization, the polarization of the light is actually rotated by 90° . This is why a $\lambda/2$ plate is also known synonymously as a “polarization rotator”.

When we talk about the “0°” and “90°” settings from now on (later “-45°” and “45°”), this angle always refers to the *rotation angle of the polarization* and never the rotation angle of the $\lambda/2$ plate.

A sketch describing how the $\lambda/2$ plate operates is shown in the diagram to the right. Light incident on the wave plate is altered such that polarization components not aligned with the fast axis of the birefringent crystal are retarded. For linearly polarized light, the result is that the polarization is rotated by a value twice as large as the rotation of the $\lambda/2$ plate.

The receiving unit “Bob” consists of a polarizing beamsplitter cube and two detectors. The polarizing beamsplitter cube reflects the vertically-polarized (90°) component of the incident light, while passing the horizontally-polarized (0°) component, as seen in the diagram to the right.

If the polarization state of the light sent by Alice is set to 0° , the photon will pass through the beamsplitter (designated as event “0”). If the wave plate is set to rotate the polarization by 90° , the photon will be reflected by the beamsplitter (designated as event “1”).



5.3.2. Key Distribution – The Right Way

While the method with one basis (0° or 90°) is sufficient to transfer data from Alice to Bob, it is not able to guarantee safety from interception. A second basis comes into play to accomplish this. In addition to the basis with 0° and 90° , which we will now call the “+ basis”, a second basis with -45° and 45° is used. From here on we will call it the “x basis”.

Now the setup looks like Figure 3. This is also the setup used for quantum cryptography and for this experiment package.

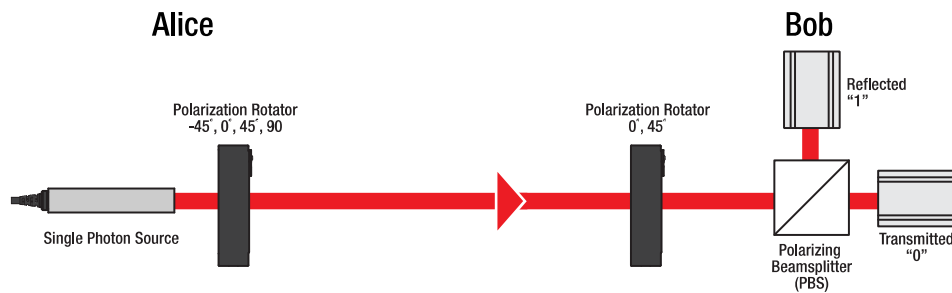


Figure 3 Quantum Cryptography Setup with the Bases + (0° and 90°) and x (-45° and 45°)

Now Alice has to make two *random* decisions for key generation:

- Alice has to select her basis at random, + or x
- Alice has to select a random bit, 0 or 1
 - Selecting 0 with the + basis means the setting 0°
 - Selecting 1 with the + basis means the setting 90°
 - Selecting 0 with the x basis means the setting -45°
 - Selecting 1 with the x basis means the setting 45°

Bob sets his polarization rotator to differentiate between the + and x bases. Accordingly Bob only needs the settings 0° and 45° .

If Bob selected the + basis and Alice sends in the + basis, Bob obtains an unambiguous result; this applies correspondingly if both choose the x basis. But what if Bob chooses a different basis than Alice? The result of choosing a different basis than Alice is that 45° polarized light will be sent to the beamsplitter. For a continuous beam, half is transmitted and half is reflected. However, assuming that only one photon is sent, only one of the two detectors can respond. The detector that responds is then left to chance. If the two bases do not match, Bob will nevertheless measure a signal on one of the two detectors. The probability of detecting the photon on one of the two detectors is 50% respectively.

In the table that follows, the different cases are shown again as an overview:¹

Alice			Bob				Same basis?
Basis	Bit	Angle	Basis	Angle	Detector "0"	Detector "1"	
+	0	0°	+	0°	100%	0%	Yes
+	1	90°	+	0°	0%	100%	Yes
x	1	45°	+	0°	50%	50%	No
x	0	-45°	+	0°	50%	50%	No
+	0	0°	x	45°	50%	50%	No
+	1	90°	x	45°	50%	50%	No
x	1	45°	x	45°	0%	100%	Yes
x	0	-45°	x	45°	100%	0%	Yes

If Alice were to send a signal comprising of random bits in random bases and Bob analyzed the signal using a random basis – how does this become the key for data transmission?

The answer is that both Alice and Bob will tell each other which BASIS is being used to transmit each bit at a later time. In the last three columns of the table, the result is only unambiguous (100%) when the bases are the same.

In practice, Alice and Bob will go through each of the measurements and only communicate "+" or "x". When the two are different, both discard the measurement. But if both bases are the same, then BOTH know which BIT was transmitted based on the result obtained by Bob's detectors. The bases, not the bits, are ever communicated publicly. Therefore the encryption key is derived from the measurements in which Alice and Bob chose the same basis.

As soon as Alice and Bob have gone through all measurements this way, both are in possession of the (random) key. Now Alice can encrypt the message and send it in the + basis. Bob receives the message in the + basis and is then able to decrypt it.

Section 5.4 expands on this protocol and introduces an eavesdropper into the process. The results of the same exercise have much different results, which allow Alice and Bob to detect the presence of this eavesdropper.

¹ If slight variations of the table should be found in other implementations of this experiment, this is probably caused by a different polarization of the incident laser. If it is vertical, then the 0° (Alice) and 0° (Bob) make a digital 1.

5.4. Detection of an Eavesdropper

Let us examine the situation of an eavesdropper “Eve” placed between Alice and Bob. Eve consists of the same components as Alice and Bob, only in the reverse sequence. This is shown in Figure 4.

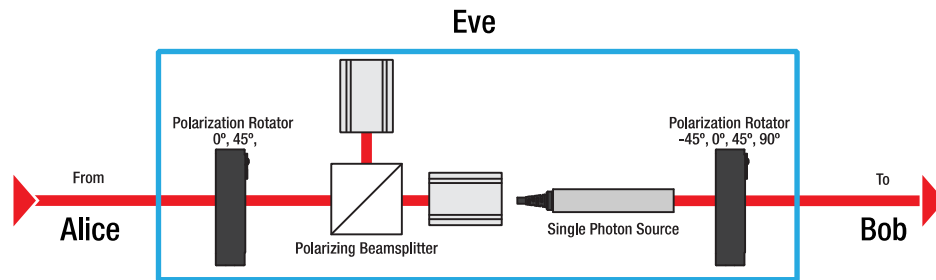


Figure 4 Eavesdropper Eve between Alice and Bob

Eve is placed in a position to measure the light coming from Alice and then attempt to transmit the identical information to Bob. Consider the following two possibilities:

- Eve chooses the same basis as Alice: In this case, Eve measures the signal that Alice sends correctly. Therefore, Eve will transmit the correct result to Bob in the same basis that was initially sent by Alice. Now Bob randomly chooses his basis, and once again there are two possibilities:
 - Bob chooses the same basis as Alice: Eve has transmitted the signal correctly using the same basis. Thus Bob obtains exactly the polarization state sent by Alice without detecting the presence of Eve.
 - Bob chooses the other basis: The basis used to receive the signal is different than the basis of the signal transmitted by Eve. Therefore, one of his detectors will respond at random. However, when Alice and Bob now compare their bases (same result as in the preceding subsection), this measurement is discarded anyway because of the different bases used by Alice and Bob.
- Eve chooses the wrong basis: In this case, Eve chooses a basis that differs from the one used by Alice and one of Eve's detectors will respond at random. Therefore, Eve is not able to judge whether she chose the correct basis. When Eve sends her signal to Bob, she will send the bit in the same basis that she received Alice's signal with.

Because Bob is also randomly choosing his basis, there are two possibilities:

- Bob chooses a different basis than Alice: This measurement is discarded after Alice and Bob compare bases.

- **Bob chooses the same basis as Alice:** This case produces the error which allows Alice and Bob to detect Eve eavesdropping. Keep in mind that Alice and Bob have confirmed that they sent and received the signal using the same basis, so the measurement is not discarded. However, Eve was eavesdropping in a different basis. This means two random detections took place: Eve in intercepting Alice's signal (because her basis did not match Alice's) and Bob's in receiving the intercepted signal (because his basis did not match Eve's).

In half of the cases the correct detector responds for Bob, so that he receives the same bit which Alice sent. But in the other half of the cases, the other detector will detect the photon. Therefore Bob obtains a different bit than the one sent by Alice.

To summarize: This represents a case where Alice and Bob obtain *different bits with the same bases* (which can never happen without third party interference). Therefore, the test for a spy is simple. After Alice and Bob have compared bases, they choose a certain number of bits with matching bases to compare publicly. If these test bits are identical, then there was no eavesdropper in the system.² But if errors are found in approximately 25% of the compared bits, then the communication was intercepted.

Although it may appear that Eve is only discovered only after eavesdropping, this is not the case since only one test encryption key has been generated so far. Even if Eve was eavesdropping (and therefore intercepted a certain number of bits undetected), this is inconsequential since no part of the actual message has been encoded or transmitted.

The individual cases are presented again briefly in a table for an overview. Only the cases where Alice and Bob use the same bases are considered here. The remaining measurements are discarded during the basis comparison process.³

Basis used by Alice and Bob	Basis used by Eve	Error?	Bits match for Alice and Bob
+	+	No	100%
+	x	In part	50%
x	+	In part	50%
x	x	No	100%

² Statistically it is possible for all test bits to be randomly correct. Therefore the sample size of bits must be large enough to ensure that the result is statistically significant.

³ The 25% error rate can be calculated from the table. If Alice and Bob choose the + basis, then Eve also chooses the + basis in 50% of all cases, which cannot be detected. But in 50% of all cases she chooses the x basis. Furthermore, in 50% of these cases the correct detector responds due to the random chance associated with an incorrect signal transmission. Therefore, the total error rate is $50\% \times 50\% = 25\%$.

5.5. What Does “Random” Mean?

As described in Section 5.2, the one-time pad requires the completely random selection of the encryption key. This means computer-generated, pseudo-random numbers are not a solution for 100% security. However, quantum physics offers numerous possibilities for true randomness. For example, a photon that hits a 50:50 non-polarizing beamsplitter is transmitted or reflected entirely at random. Half are transmitted and the other half reflected on average, but the “decision” of an individual photon is completely random. While this particular principle applies to photons, many other processes such as radioactive decay are also entirely random.

In practice, we can interpret a photon reflected by the beamsplitter as a binary 0 and a photon transmitted through the beamsplitter as a binary 1. In the case of conventional light incident on two single photodetectors after a beamsplitter, if the intensity on the detectors is the same, then the distribution of hits on the detectors can be considered completely random as well.

Particularly, quantum physics random number generators are a key component of quantum cryptography data networks. Some commercially available options already exist: <http://www.idquantique.com/random-number-generation/>.

5.6. What Prevents One from Simply Copying the Information?

Consider the scenario where Eve could simply copy the photon carrying the information. In this case, the security of quantum cryptography would be wiped out, since she could send the original photon on to Bob and carry out her measurement on the copied photon. Eve could then intercept the key bits without Alice and Bob detecting the eavesdropping.

However, the exact copying of a quantum physics state is actually impossible. This principle is known as the “no-cloning theorem” which was formulated and proven in 1982. In general, this theorem states that no quantum state can be exactly copied without altering its state. Therefore, Eve could not copy a photon from Alice without altering it, and cannot send an unaltered photon to Bob while keeping a copy for analysis.

5.7. Experimental Procedure

The sequence of steps for this experiment are taken from the the BB84 protocol. You can find the original publication here:

<http://researcher.watson.ibm.com/researcher/files/us-bennetc/BB84highest.pdf>

The sequence of the BB84 protocol experiment is:

1. Key transmission	Alice chooses a random basis (x or +) and a bit (0 or 1). Bob then randomly chooses his basis (x or +). Both set their wave plates accordingly. Then the photon is sent through the setup (with us, the laser pulse). Bob notes whether he measured a 0 or a 1. Repeat this step many times.
2. Erasing incorrect bases	Alice and Bob go through their measurements and tell each other which bases they chose. They keep the bits measured when bases used by both Alice and Bob were the same. Any others are discarded from the sequence. Note: Only the bases are communicated between Alice and Bob (this can even be done publically). After comparison, the remaining bits become the secret encryption key between Alice and Bob.
3. Testing for a spy	Alice and Bob publically compare a sample of the transmitted bits in step 2. If errors exist, this confirms the presence of a spy and the key is erased.
but omitting Error reconciliation, Privacy amplification, and Message authentication	However, if no eavesdropper is detected, the test bits are removed and the remaining bits are the final encryption key.
4. Encrypting the message	Using the key generated in step 3, Alice can encrypt the message.
5. Transmitting the message	Alice sends the encrypted message to Bob. This is done publicly.
6. Decrypting the message	Bob decrypts the message with the help of the generated encryption key.

There are additional steps in the protocol, which are not implemented for this experiment:

- Authentication: A few bits are exchanged at the start of communication according to a key established in step 1 to authenticate that the messages in step 2 and 3 to avoid man-in-the-middle attacks. If no errors occur then this confirms there is no eavesdropper in the line at the outset. One way to accomplish this step is to save a few bits from a previous communication for authentication in the next communication.

Error correction: Since no system is perfect and measuring errors always occur, certain algorithms are used for error correction. These algorithms are not discussed here as it is outside the scope of the experiment.

5.8. Classical Light versus Single Photons

The one major difference between this analogous experiment and a true quantum cryptography setup is that genuine safety from interception can only be guaranteed if a single photon source is used. This means the information of a bit has to be transported only by a single photon, since according to Section 5.6 it cannot be copied or measured without changes.

If any classical light source (even weakened lasers) is used instead of a single photon source, then Eve cannot be detected. All that is required to eavesdrop is for Eve to separate a portion of the transmitted light for detection/analysis while sending the remainder to Bob unnoticed.

Because the experiment detailed in this kit uses a pulsed light source (still not a single photon source), it cannot truly be employed as a perfect encryption system. However, the sequence of the protocol is completely identical to a true quantum encryption system.

5.9. Entanglement

A question often arises about the relationship between quantum cryptography and quantum entanglement. It is important to note that the BB84 protocol does not require polarization-entangled photons. This becomes clear upon recognizing the first papers on quantum cryptography with entangled photons were published after 1984. For reference:

- A.K. Ekert, Phys. Rev. Lett. **67**, 661 (1991)
- C. H. Bennett, G. Brassard, N. D. Mermin, Phys. Rev. Lett. **68**, 557 (1992)

Moreover, the field of quantum cryptography is very dynamic and has since seen substantial improvements and introduced new protocols. These are more complex but do not require single photons. These are what are referred to as “decoy states.” For reference:

- W.-Y. Hwang, Phys. Rev. Lett. **91**, 057901 (2003)
- H.-K. Lo, X. Ma, K. Chen, Phys. Rev. Lett. **94**, 230504 (2005)

Chapter 6 Examples

6.1. Encryption Protocol without Eve (Two Letters)

Step 1: Alice and Bob randomly select their bases, and Alice also selects her bits

Alice

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
Basis	x	x	+	x	+	+	+	x	x	+	x	x	x	+	+	x	+	x
Bit	1	0	0	1	1	0	0	1	1	1	0	1	0	0	0	1	0	1

Bob

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
Basis	+	x	x	x	+	+	+	x	+	x	x	+	+	x	+	+	+	x
Bit																		

Step 2: Alice sends the bits in the chosen basis and Bob records the bits he measures. During Bob's measurements, he uses the bases chosen randomly by him. Bob will then interpret Alice's transmission as in the example below. Please note that in cases where the bases do not match, a 0 or 1 bit result is selected at random.

Bob

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
Basis	+	x	x	x	+	+	+	x	+	x	x	+	+	x	+	+	+	x
Bit	1	0	1	1	1	0	0	1	1	1	0	1	0	1	0	0	0	1

Step 3: Alice and Bob exchange their bases ("I have +" or "I have x"). Both Alice and Bob will note which of the transmitted bits were sent using the same basis (see below):

Alice

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
Basis	x	x	+	x	+	+	+	x	x	+	x	x	x	+	+	x	+	x
Bit	1	0	0	1	1	0	0	1	1	1	0	1	0	0	0	1	0	1

Bob

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
Basis	+	x	x	x	+	+	+	x	+	x	x	+	+	x	+	+	+	x
Bit	1	0	1	1	1	0	0	1	1	1	0	1	0	1	0	0	0	1

The resulting key generated by the two of them is: "0 1 1 0 0 1 0 0 0 1". Both know this key even though they have only exchanged their bases.

Step 4: Alice encrypts two letters with the key that was just generated. Read the binary representation of Q and M from the alphabet table, then perform binary addition on the first and second rows, see Chapter 9 for reference tables.

Letter	Q					M				
Data Bit	1	0	0	0	0	0	1	1	0	0
Key Bit	0	1	1	0	0	1	0	0	0	1
Encrypted Bit	1	1	1	0	0	1	1	1	0	1

Step 5a: Alice sends the encrypted message using the + basis (0° to indicate 0 and 90° to indicate 1). She therefore selects the following angle settings in the sequence below:

90° , 90° , 90° , 0° , 0° , 90° , 90° , 90° , 0° , 90°

Step 5b: Bob receives Alice's transmission using the + basis as well (reflected light = 1, transmitted light = 0). Bob will record the following received bits:

Received Bit	1	1	1	0	0	1	1	1	0	1
--------------	---	---	---	---	---	---	---	---	---	---

Step 6: Bob then uses the encryption key to decode the message (binary addition of the first and second lines)

Received Bit	1	1	1	0	0	1	1	1	0	1
Key Bit	0	1	1	0	0	1	0	0	0	1
Data Bit	1	0	0	0	0	0	1	1	0	0
Letter	Q					M				

6.2. Encryption Protocol with Eve

The process of generating the key is discussed again here, but this time with Eve who is eavesdropping. Her presence is discovered by comparing test bits.

Step 1: Alice, Bob and Eve randomly select their bases, and Alice also randomly selects bits to send.

Alice

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
Basis	x	x	+	x	+	+	+	x	x	+	x	x	x	+	+	x	+	x
Bit	1	0	0	1	1	0	0	1	1	1	0	1	0	0	0	1	0	1

Bob

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
Basis	+	x	x	x	+	+	+	x	+	x	x	+	+	x	+	+	+	x
Bit																		

Eve

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
Basis	x	x	+	+	x	+	+	+	x	x	x	x	+	x	+	x	+	+

Step 2: Alice then sends the bits using her selected basis and Bob records the bits he measures. However, in this scenario, Eve is between Alice and Bob and selects her basis randomly as well (either 0° and 45°). If Eve's basis matches the one chosen by Alice, Eve will transmit the correct bit. If Eve selected the incorrect basis, she will transmit a random bit based on her interpretation of Alice's signal (0 or 1). Bob, however will record the bit that he receives from Eve. The measurements performed by Eve and Bob will appear as in the table below (Eve's measurements are shown for explanatory reasons, but are generally hidden from both Alice and Bob⁷):

Eve

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
Basis	x	x	+	+	x	+	+	+	x	x	x	x	+	x	+	x	+	+
Bit	1	0	0	0	1	0	0	1	1	0	0	1	1	1	0	1	0	1

Bob

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
Basis	+	x	x	x	+	+	+	x	+	x	x	+	+	x	+	+	+	x
Bit	1	0	1	1	0	0	0	0	1	0	0	0	1	1	0	0	0	0

In the tables below, the same set of data measurements is presented but with all events determined by random chance highlighted in green. Green measurements in the first table indicate random events produced between Alice and Eve, while green measurements in the second table indicate random events produced between Eve and Bob.

In either case, neither Eve nor Bob know whether their measurement was randomly generated.

Eve

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
Basis	x	x	+	+	x	+	+	+	x	x	x	x	+	x	+	x	+	+
Bit	1	0	0	0	1	0	0	1	1	0	0	1	1	1	0	1	0	1

Bob

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
Basis	+	x	x	x	+	+	+	x	+	x	x	+	+	x	+	+	+	x
Bit	1	0	1	1	0	0	0	0	1	0	0	0	1	1	0	0	0	0

⁷ Similar to the experiment without Eve, the measurements where the bases do not match are random.

Step 3: At this point, Alice and Bob exchange the bases used for transmitting and receiving ("I have +" or "I have x"). They highlight the measurements where the bases match.

Alice

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
Basis	x	x	+	x	+	+	+	x	x	+	x	x	x	+	+	x	+	x
Bit	1	0	0	1	1	0	0	1	1	1	0	1	0	0	0	1	0	1

Bob

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
Basis	+	x	x	x	+	+	+	x	+	x	x	+	+	x	+	+	+	x
Bit	1	0	1	1	0	0	0	0	1	0	0	0	1	1	0	0	0	0

Step 4: Now Alice and Bob compare the bits for which their bases matched (the ones circled in red)

Alice: 0 1 1 0 0 1 0 0 0 1
 Bob: 0 1 0 0 0 0 0 0 0 0

As seen above, there are differences between the two bit sequences – these are circled in blue. The errors occurred three times out of ten, which is close to the expected 25%.⁸ The presence of errors in this test string alerts Alice and Bob to the presence of an eavesdropper and the protocol would be abandoned at this point.

⁸ There is no system underlying the numbers chosen here. The fact that Bob has so many zeroes is therefore pure chance. Furthermore, the combination of "Alice=1, Bob=0" three times for the incorrect pairs is random. When the random events marked in green are chosen differently, the bits compared by Alice and Bob also change correspondingly.

Chapter 8 Experiment

The experiment is divided into three sections:

- **Section 8.1:** Generating a key with a length of at least 10 bits
- **Section 8.2:** Encryption and transmission of 2 letters
- **Section 8.3:** Installation of Eve and detection of an eavesdropper

In order to complete the exercises, the user should be familiar with the encryption principles outlined in Chapter 5. Detailed examples are presented in Chapter 6, illustrating the entire process with and without Eve. A sample measurement protocol can be found in Chapter 9.

8.1. Key Generation

Exercise 1: Set-up Alice board and use the receiver on the Eve board as the receiver Bob.

align Alice and Bob so that all 8 transmission cases produce reproducible results. During this step, the sensor electronics should be in adjustment mode (the LED indicator is yellow).

Execution: See Chapter 7 for setup instructions. A basic schematic is shown in Figure 3 and a photo of the complete setup is shown in Figure 7. The possible transmission cases are outlined in Section 7.4.

Exercise 2: Alice and Bob randomly select their bases and Alice also selects random bits for transmission. Chapter 9 provides 25-bit measuring protocol. Fill in the bits and bases that Alice will use to transmit her signal and the bases that Bob will use for detection.

Execution: This corresponds to step 1 in the example described in Section 6.1. The protocols can be found in Chapter 9.

Exercise 3: Transmit Alice's bits in the bases chosen in Exercise 2. Bob notes the bits he receives.

Execution: This corresponds to step 2 in the example described in Section 6.1. Alice and Bob have to generate a key that is at least 10 bits long. We recommend a transmission sequence of at least 25 bits for this purpose. Because of the random component in generating a key, a longer transmission is more likely to achieve the 10 matching bases for the encryption key. During this step, the sensor electronics should be in measurement mode (the LED indicator is green).

To summarize the choice of bases used in transmission, if Alice:

- Selects 0° then she sends a 0 in the "+" basis.
- Selects 90° then she sends a 1 in the "+" basis.
- Selects -45° then she sends a 0 in the "x" basis.
- Selects 45° then she sends a 1 in the "x" basis.

Alice and Bob transmit the bits according to the table prepared in Exercise 2. Alice sends her bit and Bob notes his measurement (reflected = 1, transmitted = 0).

Exercise 4: Alice and Bob publically exchange the bases used for each measurement. They then erase any measurements where the bases do not match. The remaining measurements/bits form the complete encryption key.

Execution: This corresponds to step 3 in the example detailed in Section 6.1. Alice and Bob exchange their bases ("I have chosen +” or "I have chosen x”) and mark any measurements with matching bases. The corresponding bits form the encryption key.

If 52 measurements are not enough to generate a 20 bit encryption key, then the transmission should be repeated with more measurements until a 20 bit key is generated.

8.2. Encryption and Transmission of a Four Letter Word

Exercise 5: Encrypt Alice’s message (4 letters) using the key that was generated.

Execution: The procedure is described in step 4 of Section 6.1 and also in Section 5.2.

Exercise 6: Transmit the encrypted message from Alice to Bob.

Execution: Transmission of the actual message is done entirely in one basis. Alice sends her encrypted bits (0° for 0, 90° for 1). Alice and Bob do not change the basis for data transmission. This corresponds to steps 5a and 5b in Section 6.1.

Exercise 7: Decrypt the bits received by Bob to find Alice’s message.

Execution: This process is the same as described in Section 5.2 and corresponds to step 6 in Section 6.1.

8.3. Adding Eve and Detection of Eavesdropping

Exercise 8: Place Eve between Alice and Bob, and set both sensor electronics to adjustment mode (LED lights up yellow). Adjust Eve’s receiver so that all 8 transmission cases work with Alice. Then, adjust Eve’s transmitter so that all 8 transmission cases work with Bob. Return to measurement mode for both sensor electronics (LED lights up green).

Use all three boards as intended with Eve inbetween Alice and Bob

Execution: The adjustment is performed as described in Chapter 7 (also see Figure 4 and Figure 8). The goal should be to align Eve without disturbing the setups for Alice and Bob, since the eavesdropper has no influence on the sender and receiver. Then set the sensor electronics to measurement mode.

Exercise 9: Fill out the table for Eve, which determines the random choice of the + or the x basis. Random bases are also needed Alice and Bob again, and random bits should be chosen by Alice for transmission.

Execution: There are two methods for implementing this step. In an ideal scenario where the users are isolated from each other, Eve could just choose a basis spontaneously and transmit the measured result. However, because users are in the same room, the physical act of changing a basis leads to a non-random bias for

Eve's operator. Choosing random bases at the start of the sequence prevents this bias from interfering with the results of the experiment. A sample table of bases for Eve to use can be found in Chapter 9. Keep in mind that Eve's operator does not have to record any data.

An example is also provided in Step 1 of Section 6.2.

Exercise 10: Send Alice's first bit using the basis chosen in the previous exercise. Eve chooses her first basis (randomly chosen in Exercise 9). She receives a bit which she relays to Bob using the same basis. Bob records the bit he receives. This is performed for all 52 bits/bases in the sequence.

Execution: Alice and Bob use the same procedure as in Exercise 3. Eve receives the signal from Alice using the randomly chosen basis from Exercise 9. Then she transmits her measurement (using the same basis). This exercise corresponds to step 2 of Section 6.2.

Exercise 11: Alice and Bob publically exchange their bases for each measurement. They then erase the measurements in the sequence where the bases do not match.

Execution: Similar to Exercise 4, Alice and Bob compare their bases and eliminate bits where the bases used do not match. The remainder forms the bit sequence that is used to test for an eavesdropper. This exercise corresponds to step 3 in Section 6.2.

Exercise 12: Compare Alice's and Bob's results from exercise 11.

Execution: Because of Eve's eavesdropping and transmitting bits using a random basis (not necessarily identical to Alice), the bit sequences recorded by Alice and Bob should contain errors (bits that do not match). The presence of these errors is what allows Alice and Bob to detect the presence of an eavesdropper. This is different than the result from Exercise 4 where Alice and Bob should obtain the same result. This exercise corresponds to step 4 in Section 6.2.

Chapter 9 Measuring Protocols

This section contains the measuring protocols for Alice, Bob and Eve.

In order to make them easy to print, each of them is on one page (the manual is also available as a free download at discovery.thorlabs.com).

Then you find the table with the encoding of the alphabet using 5 bits per letter.

[illegible][illegible]

Use bit 1-25 for the first experiment (without Eve) and bits 26-50 for the second experiment (with Eve)

[illegible]

Angle setting (remainder)	Basis +	Basis x
Bit 0	0°	-45°
Bit 1	90°	45°

Letter																				
Data Bit																				
Key Bit																				
Encrypted Bit																				

Data Bit = letter in binary form, 4 x 5 Bit

Measuring protocol for key generation – BOB

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
Basis (+ or x)																		
Bit (0 or 1)																		

	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36
Basis (+ or x)																		
Bit (0 or 1)																		

	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52
Basis (+ or x)																
Bit (0 or 1)																

Use bit 1-25 for the first experiment (without Eve) and bits 26-50 for the second experiment (with Eve)

Generated Key:

Reminder	transmitted	reflected
Basis + (=0°)	0	1
Basis x (=45°)	0	1

Table for decryption of the message – BOB

Received Bit																		
Key Bit																		
Data Bit																		
Letter																		

Basis selection – EVE

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
Basis (+ or x)																		

	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36
Basis (+ or x)																		

	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52
Basis (+ or x)																

Binary representation of the alphabet

A	0	0	0	0	0
B	0	0	0	0	1
C	0	0	0	1	0
D	0	0	0	1	1
E	0	0	1	0	0
F	0	0	1	0	1
G	0	0	1	1	0
H	0	0	1	1	1
I	0	1	0	0	0
J	0	1	0	0	1
K	0	1	0	1	0
L	0	1	0	1	1
M	0	1	1	0	0
N	0	1	1	0	1
O	0	1	1	1	0
P	0	1	1	1	1
Q	1	0	0	0	0
R	1	0	0	0	1
S	1	0	0	1	0
T	1	0	0	1	1
U	1	0	1	0	0
V	1	0	1	0	1
W	1	0	1	1	0
X	1	0	1	1	1
Y	1	1	0	0	0
Z	1	1	0	0	1

Binary Addition Table

0	1	0	1
+ 0	+ 0	+ 1	+ 1
= 0	= 1	= 1	= 0